# Lecture 1: Introduction

CMPS 221 – Computer Organization and Design

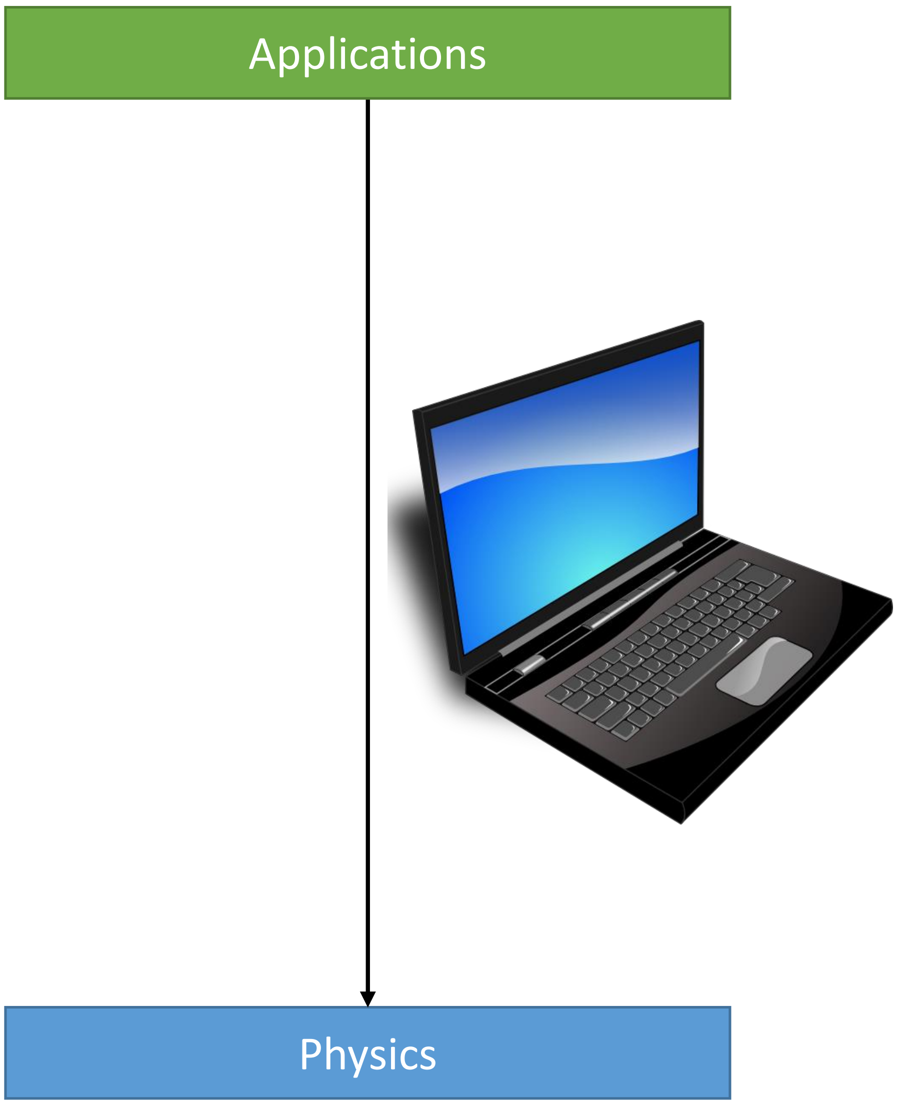
*Slides by Mahmoud Bdeir and Izzat El Hajj*

Applications

Physics

Applications

Physics




For computers, the gap is too large to bridge in one step

# Layers of Abstraction

Software

- Applications
- Tools, Methods, and Techniques
- Data Structures and Algorithms
- Programming Languages
- Compilers
- Operating Systems
- Instruction Set Architecture (ISA)
- Microarchitecture
- Logic Design
- Circuits
- Devices
- Physics

Hardware

Abstractions allow us to build complex systems and develop parts of those systems independently

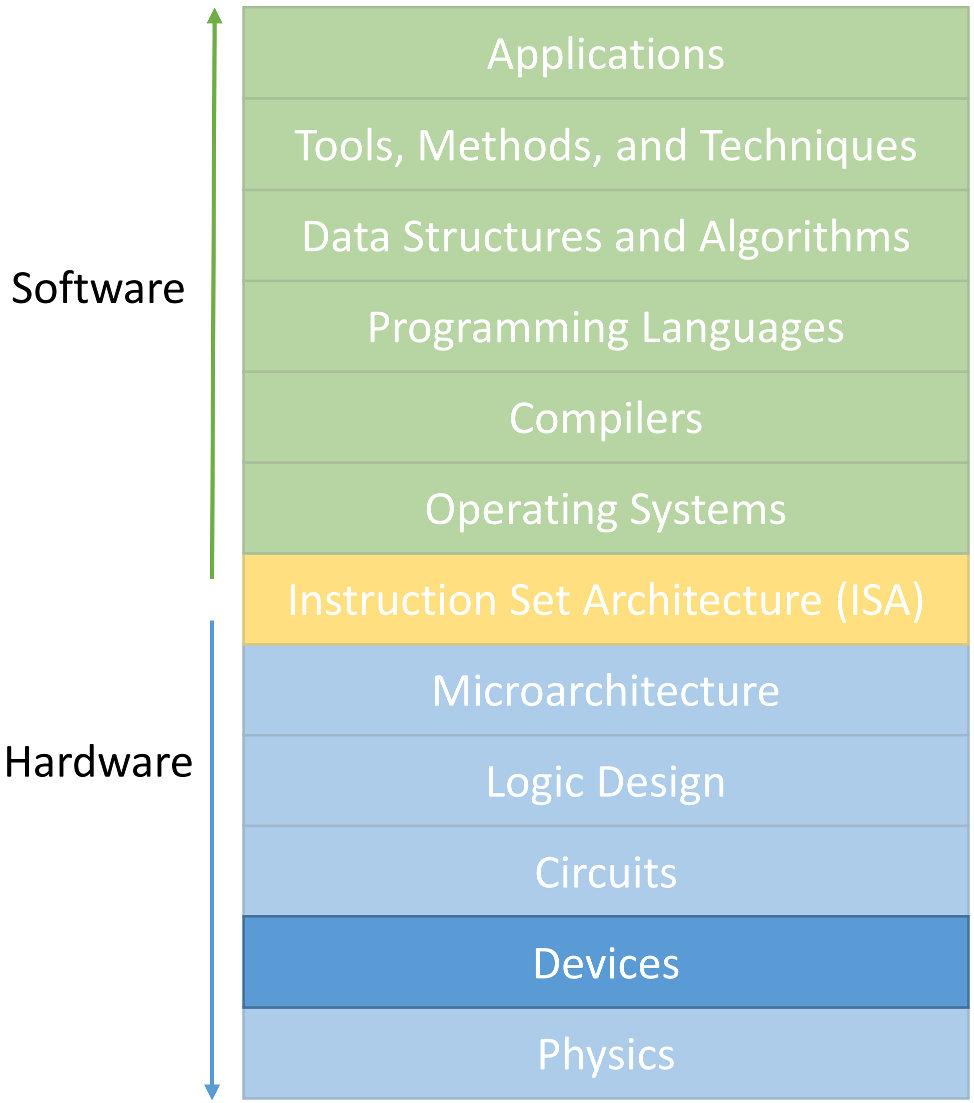
Most computer systems have the same layers of abstraction

Personal Computers

Mobile Devices

Servers and Supercomputers

# Devices

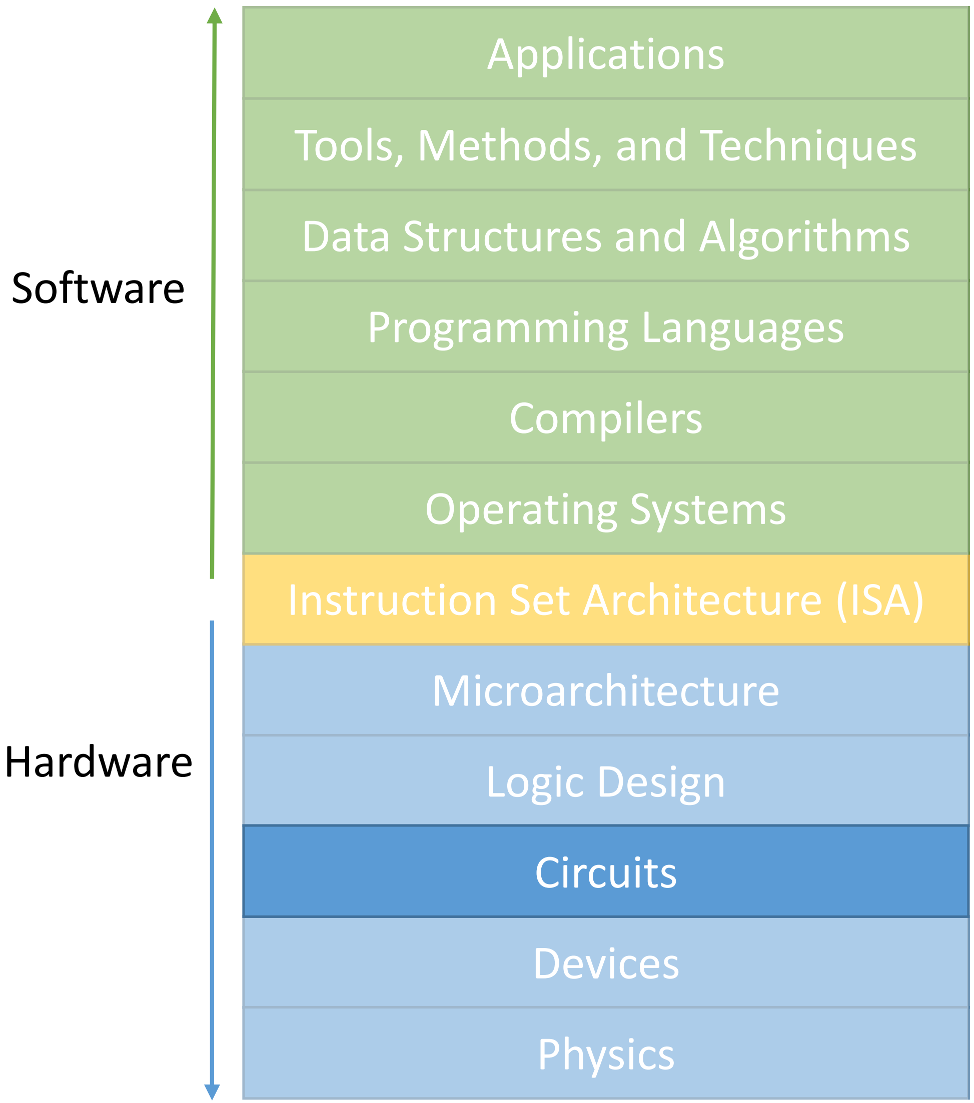



Using physical properties of materials and electrons to build devices such as resistors, inductors, capacitor, and – what computers are made of – transistors

A **transistor** functions as an ON and OFF switch controlled by electricity

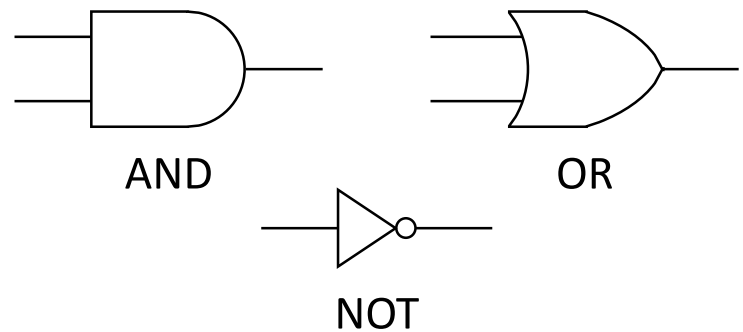
| Year | Technology | Relative performance per unit cost |
|---|---|---|
| 1951 | Vacuum Tube | 1 |
| 1965 | Transistor | 35 |
| 1975 | Integrated Circuit | 900 |
| 1995 | VLSI | 2,400,000 |
| 2013 | ULSI | 250,000,000,000 |

# Circuits



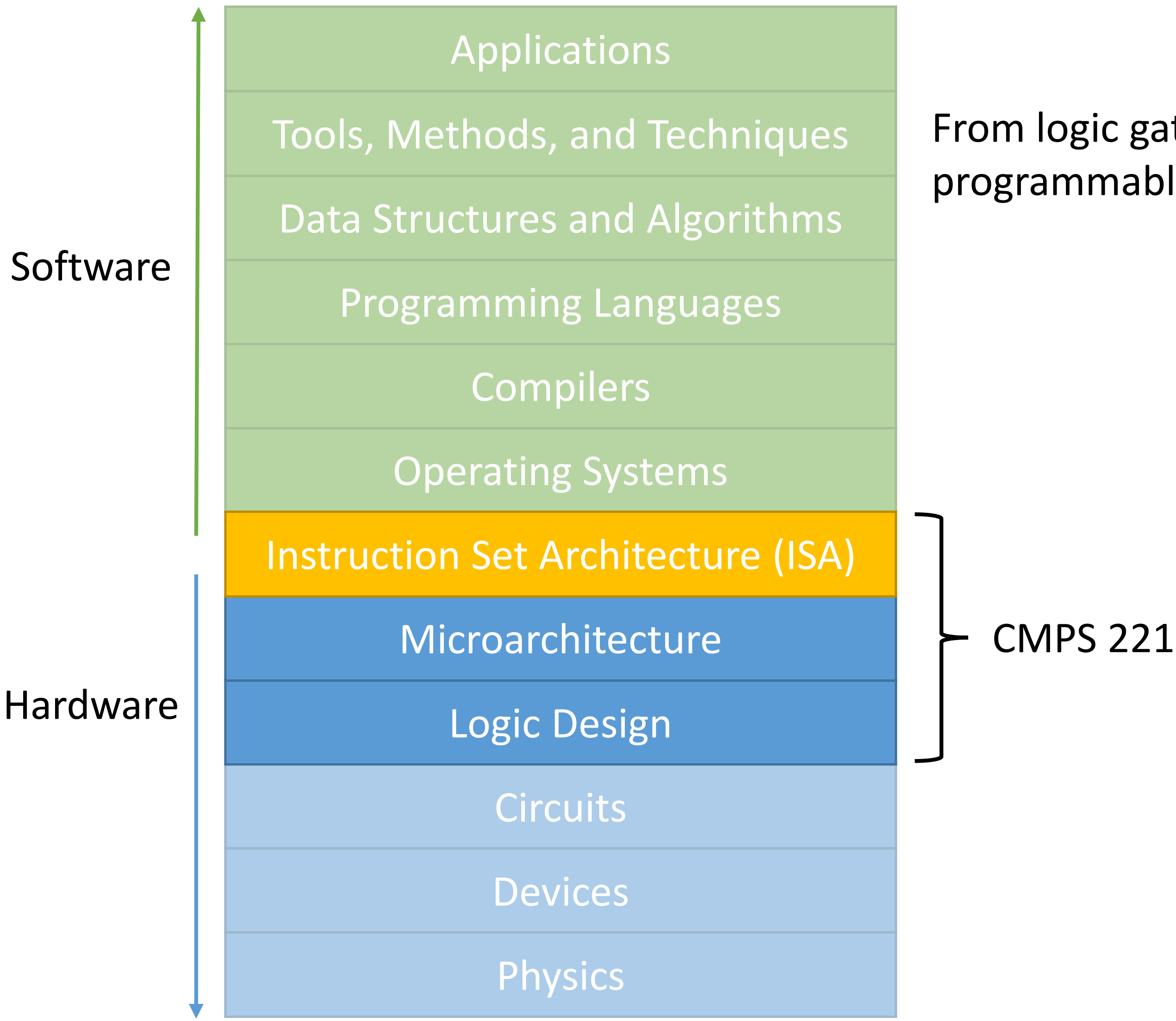

Combining transistors in different ways to build circuits with more complex functionality

Using transistors, we can build **logic gates** that perform various logic operations



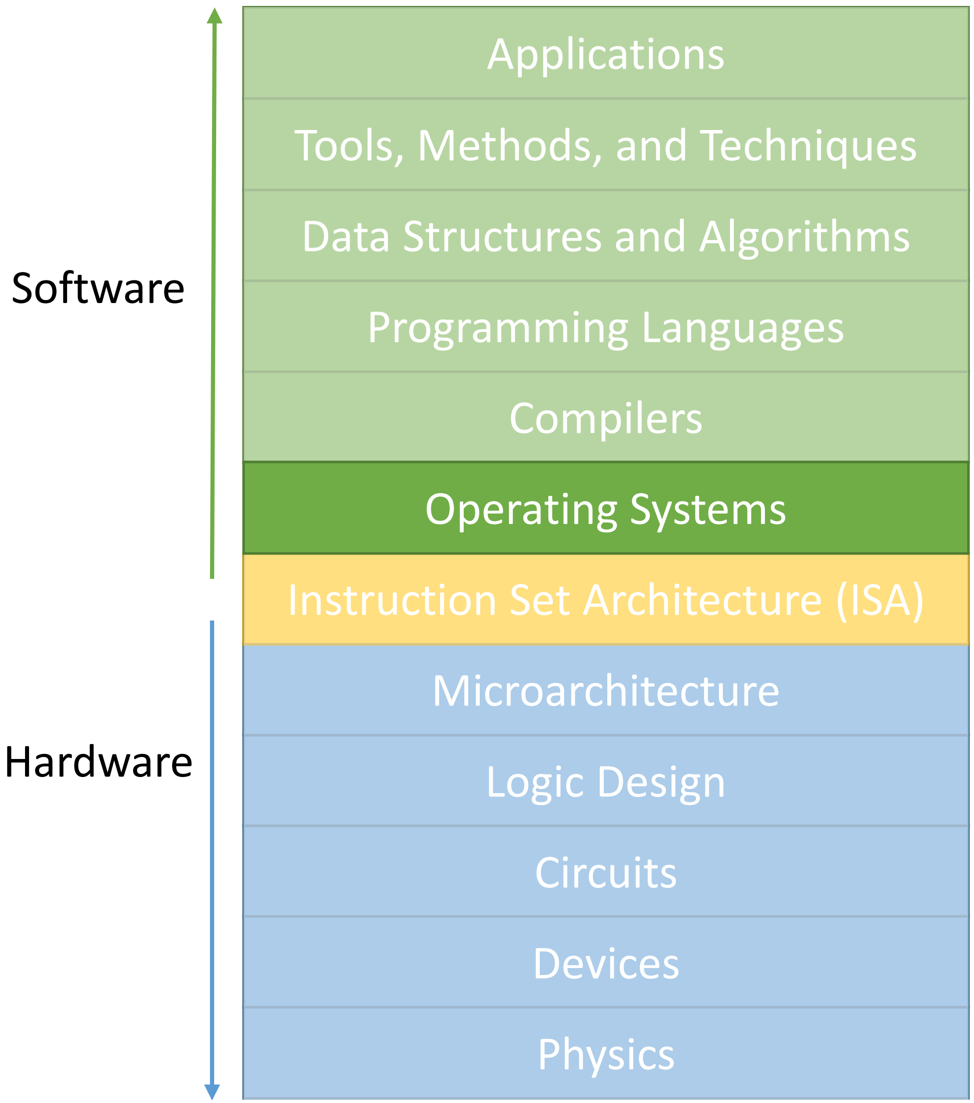

EECE 210, 310, 311, 412

# This Course




From logic gates to a general purpose programmable processor!

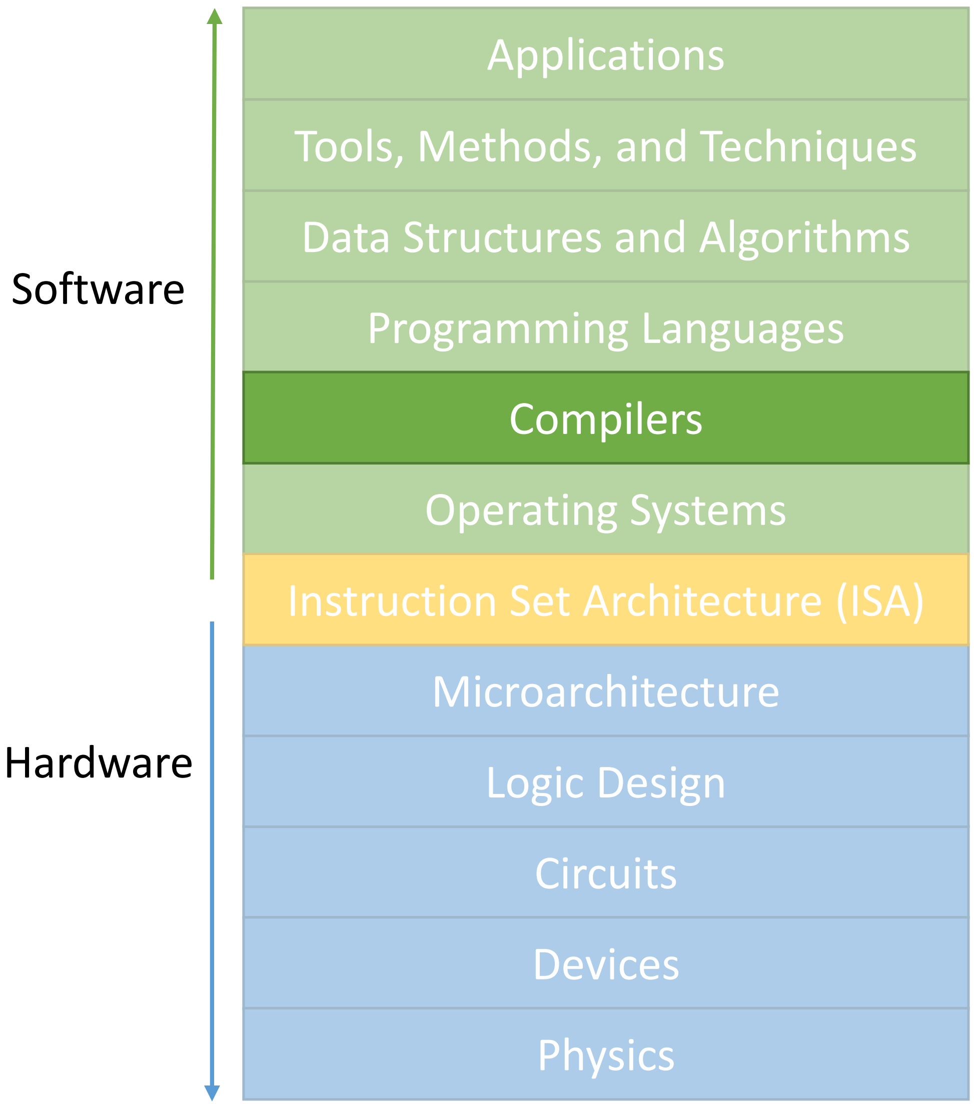
# Operating Systems




Performing a variety of services and supervisory functions such as handling input/output devices, allocating memory and storage, and protection when multiple programs are running together

CMPS 240

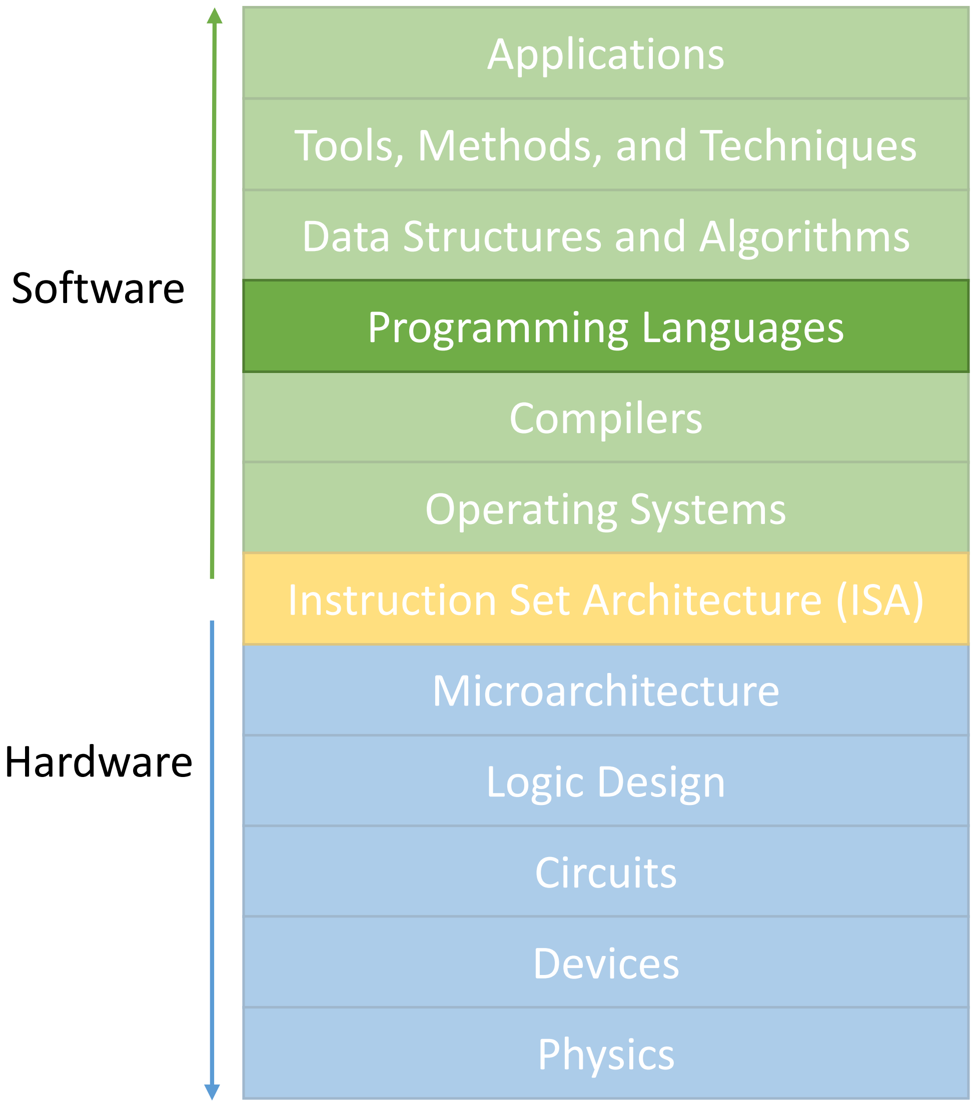
# Compilers




Translating high level code down to assembly code, applying optimizations along the way

CMPS 232

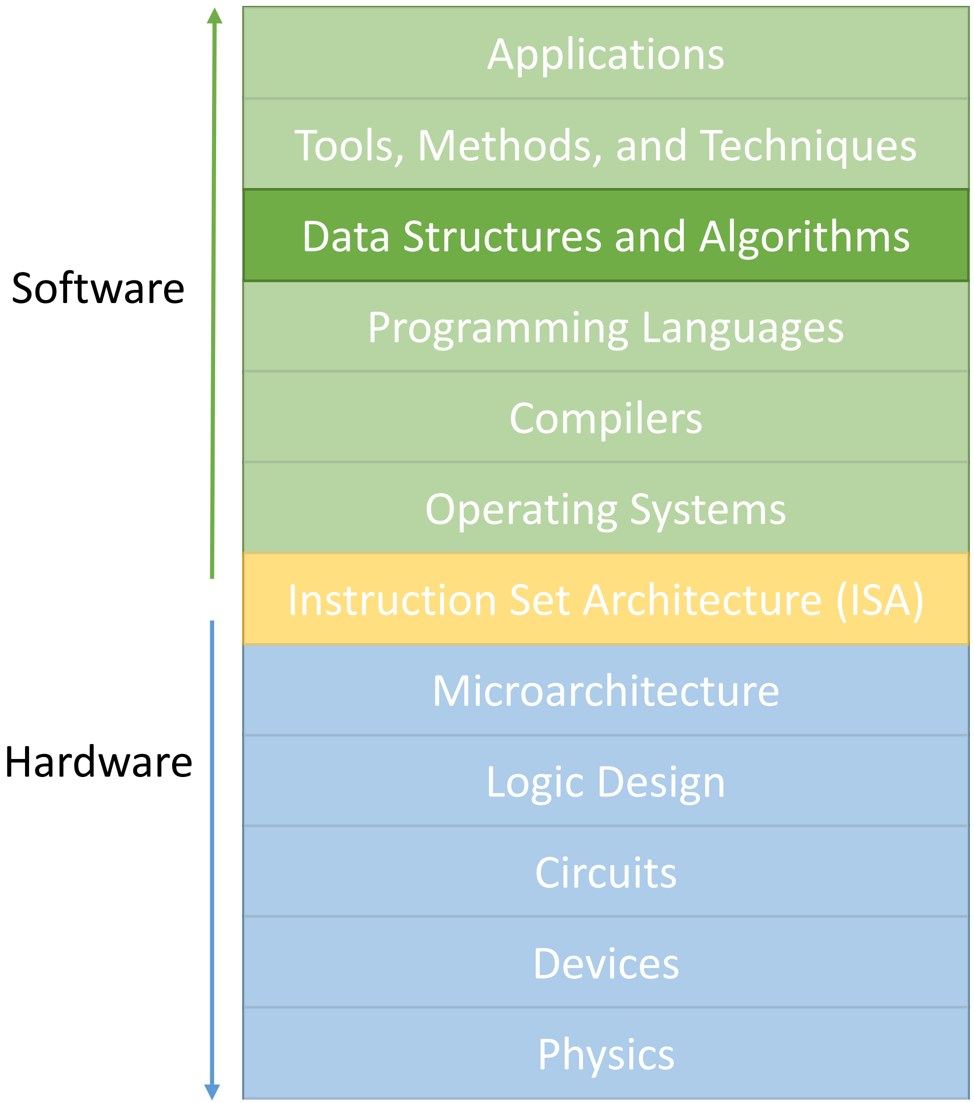
# Programming Languages




CMPS 231

Designing high-level interfaces that make it easier for programmers to write code that is concise, efficient, and safe.

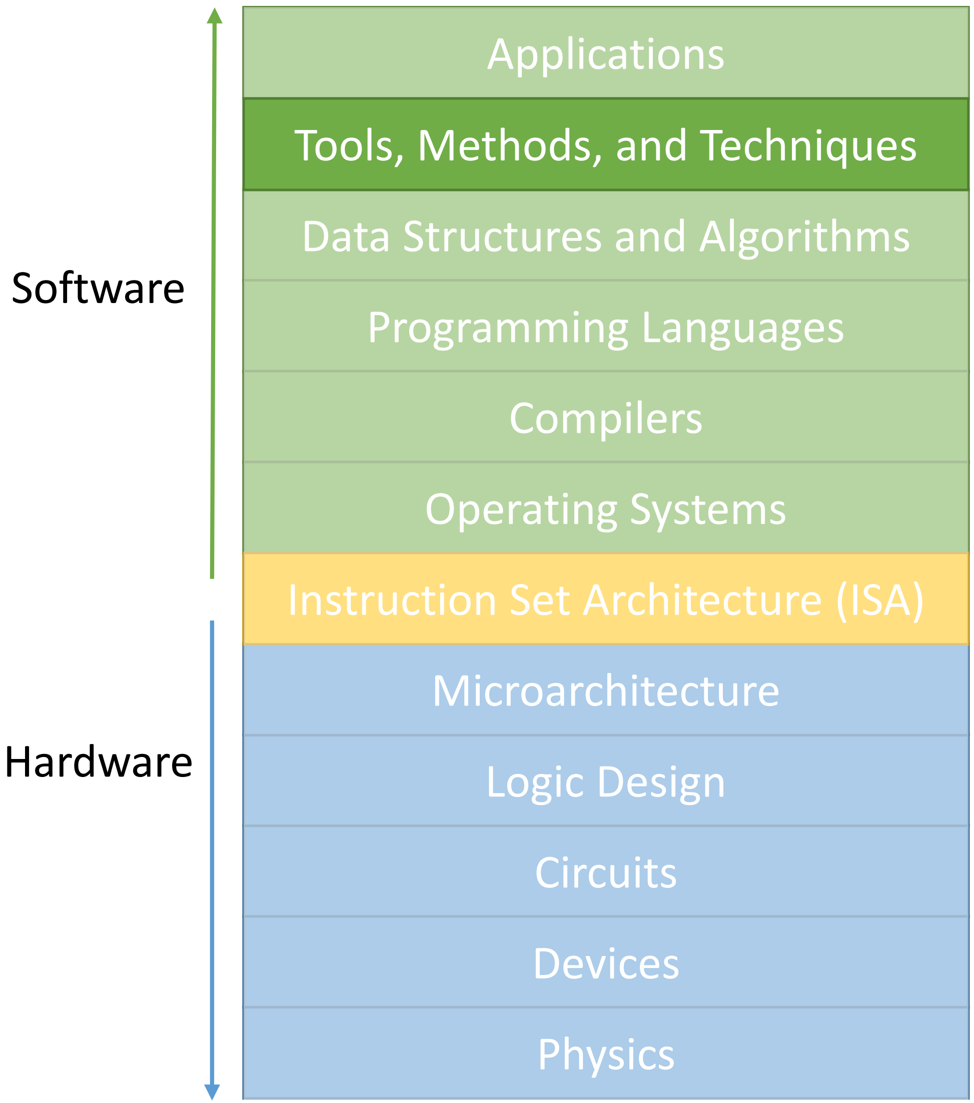
# Data Structures and Algorithms




CMPS 202, 214, 215

Creating data representations and processing techniques for performing fundamental tasks efficiently

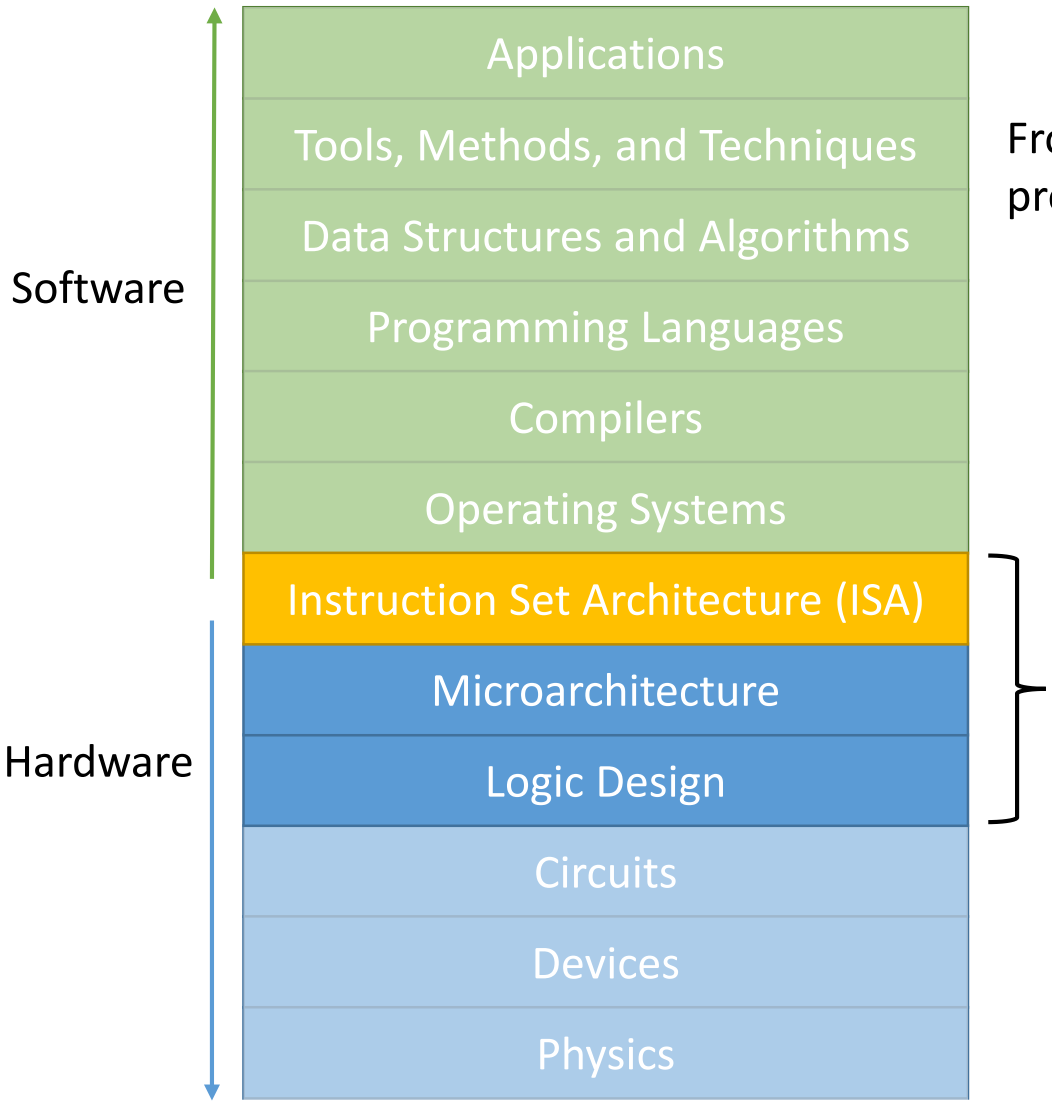
# Tools, Methods, and Techniques




CMPS 270, 271, 278

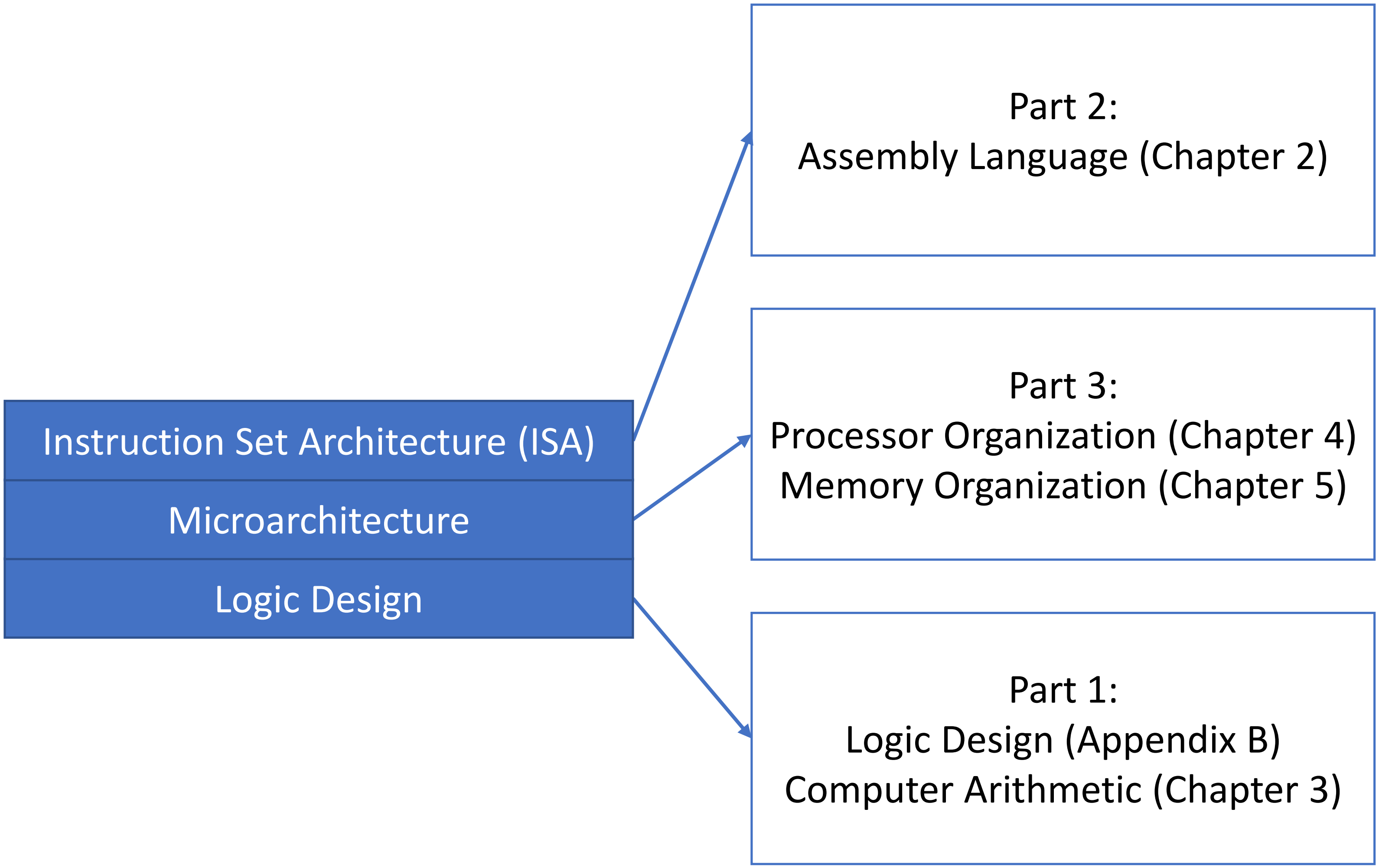
Using best practices to build software including planning, requirements, software patterns, development lifecycle, quality assurance, and build process

# Back to This Course




From logic gates to a general purpose programmable processor!

# Course Organization



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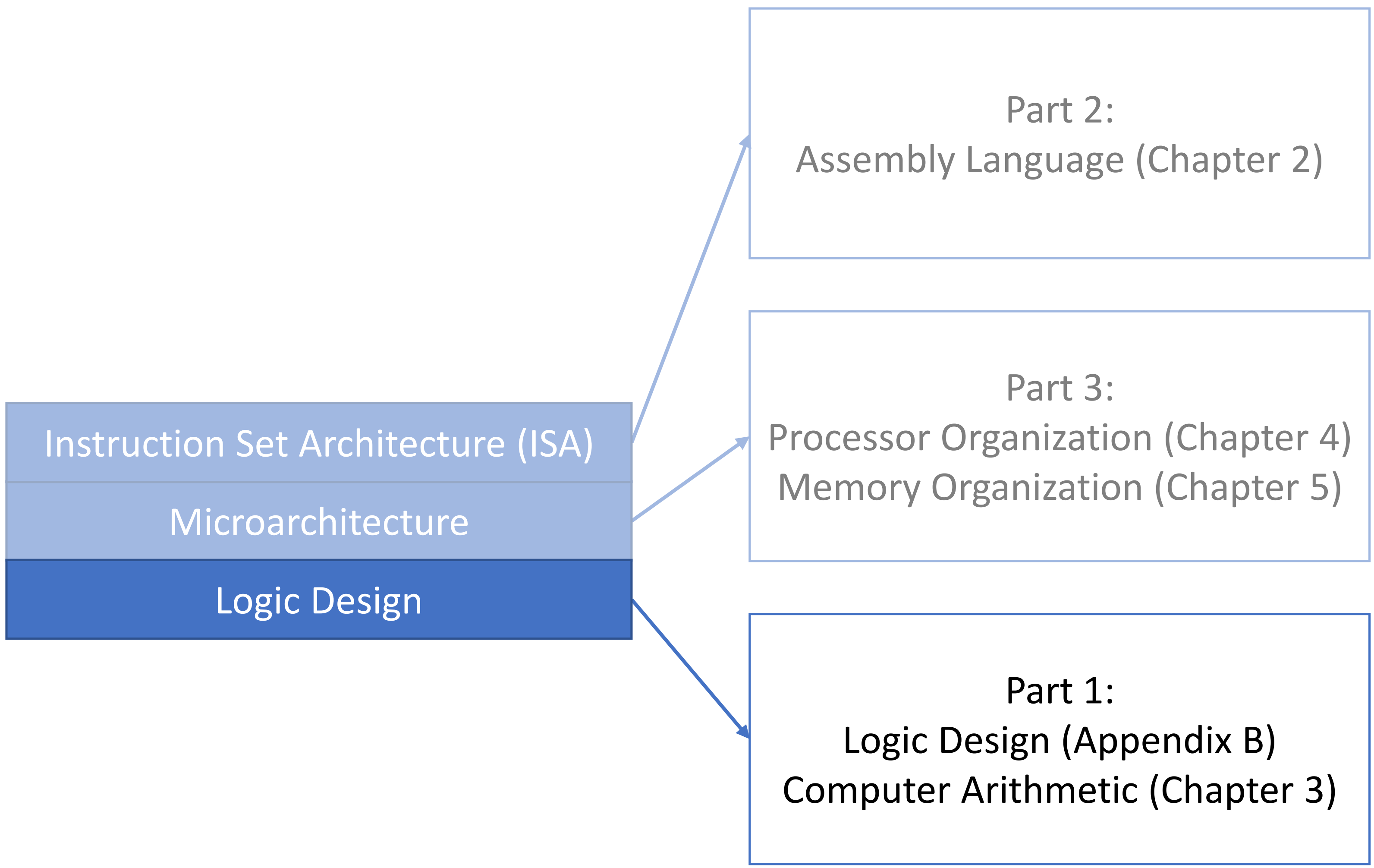
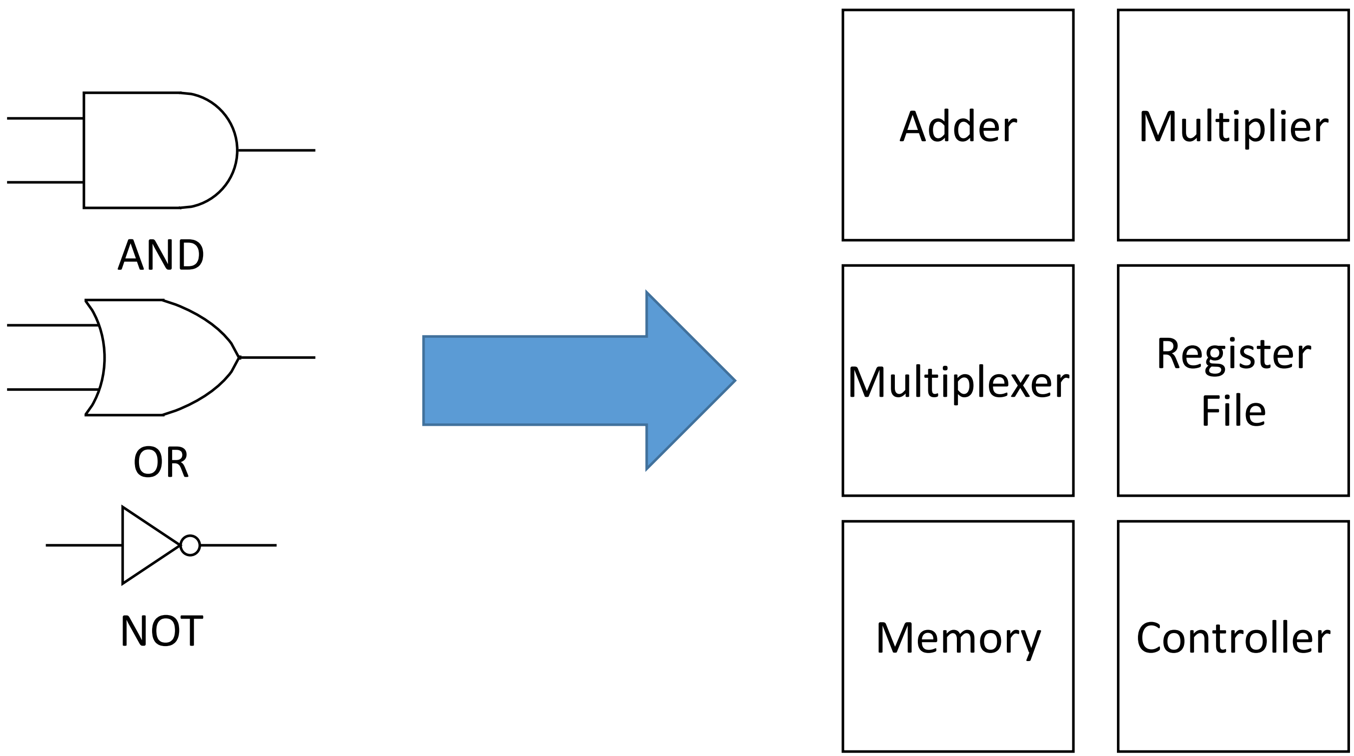
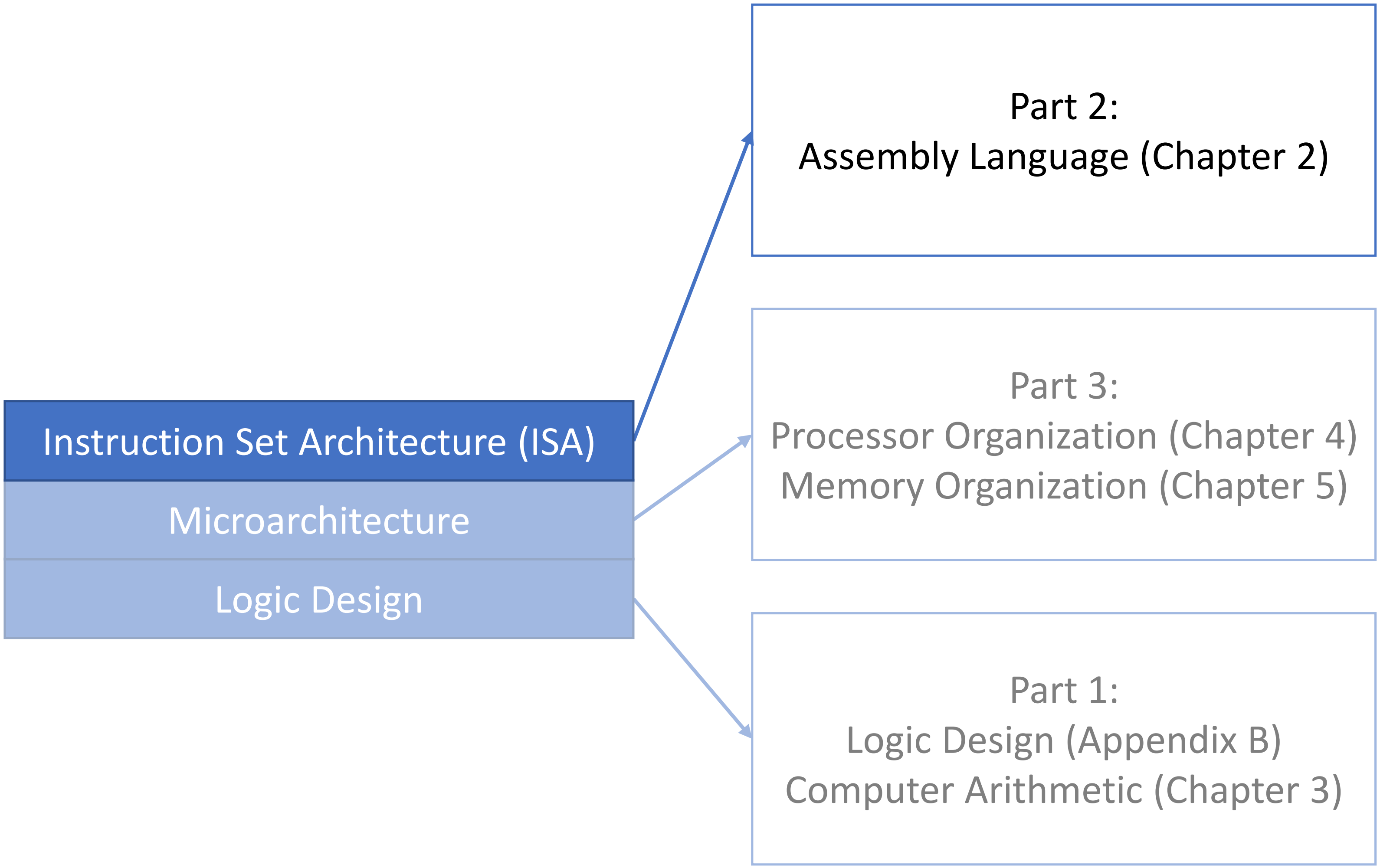


# Logic Design

Combining gates to design logic blocks that can perform arithmetic and other complex functionality



# Course Organization



# Assembly Language

```
swap (int v[], int k) {
    int temp;
    temp = v[k];
    v[k] = v[k+1];
    v[k+1] = temp;
}
```

**High-level Language Program**
(what the programmer writes)

Compiler

```
swap:  sll $2, $5, 2
       add $2, $4, $2
       lw  $15, 0($2)
       lw  $16, 4($2)
       sw  $16, 0($2)
       sw  $15, 4($2)
       jr  $31
```

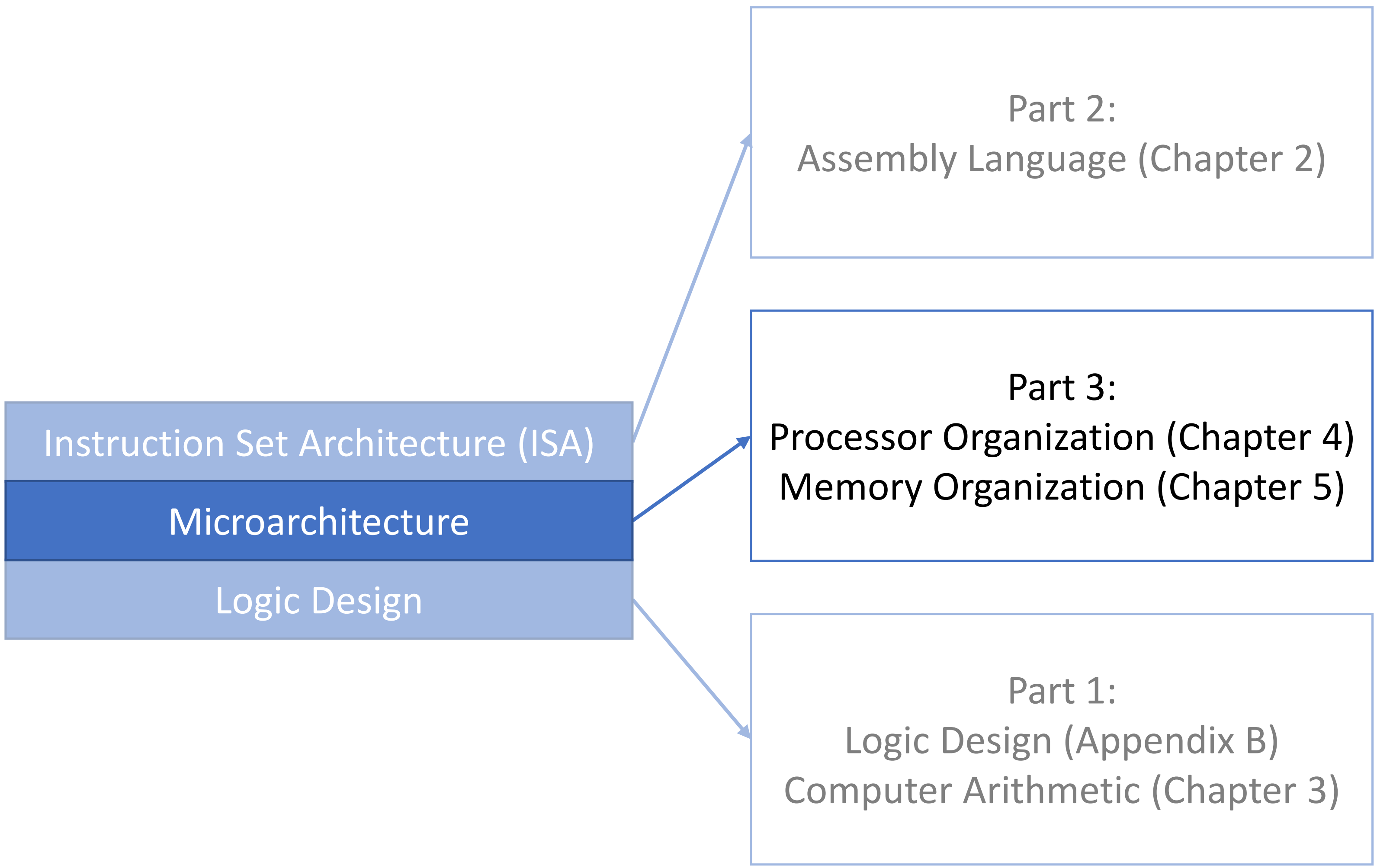
**Assembly Language Program**
(what the hardware understands)

Assembler

```
000000 00000 00101 0001000010000000
000000 00100 00010 0001000000100000
...
```

**Machine (Object) Code**
(what the hardware *really* understands)

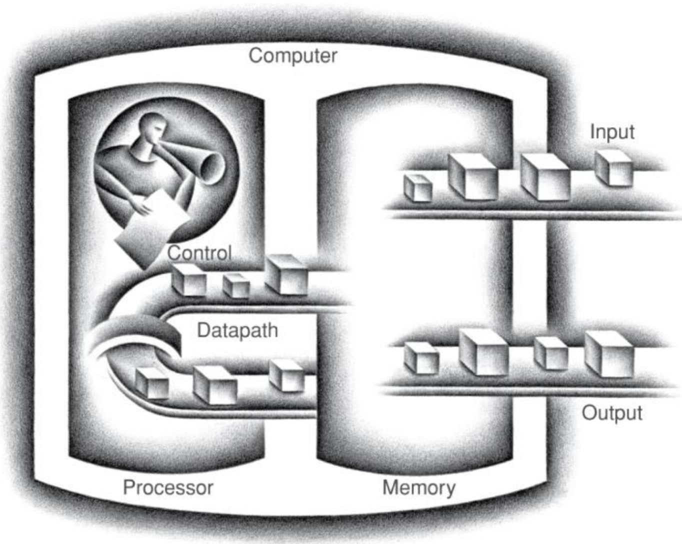
# Course Organization



# Processor and Memory Organization

Designing hardware from logic blocks that can execute a given Instruction Set

| | |
|---|---|
| Adder | Multiplier |
| Multiplexer | Register File |
| Memory | Controller |




```
sll $2, $5, 2
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```

# Textbook Sections

- Some of the content in these slides corresponds to:
  - Textbook:
    - *Computer Organization and Design, 5th Edition by David Patterson and John Hennessy, Morgan Kaufmann, 2014.*
  - Sections:
    - 1.1, 1.3

# Syllabus Overview

# Logistics

- Instructors: Hussam Remlaoui, Izzat El Hajj
- Teaching assistants: TBD
- Lectures:
  - MWF @ 8:00am-8:50am
  - MWF @ 9:00am-9:50am
  - MWF @ 10:00am-10:50am
- Office hours:
  - Posted on Moodle

# References

- Textbook
  - Computer Organization and Design (MIPS edition), by David Patterson and John Hennessy, Morgan Kaufmann.
    - $5^{th}$ edition (2014) or $6^{th}$ edition (2021) are both acceptable

- Electronic Resources:
  - Lecture slides, recordings, assignments, exams, and other materials will be posted on Moodle
  - All announcements and Q&A will be handled via Piazza (please do not send emails)

# Evaluation

- Breakdown:
  - Assignments: 15% Weekly
  - Exam 1: 25% Date posted on Moodle
  - Exam 2: 25% Date posted on Moodle
  - Final: 35% Date TBD by the registrar

- Note: Each course component assesses a different learning outcome and students are expected to meet all learning outcomes.
  - There will be **no dropping of the lowest** assignment grade
  - There will be **no shifting of percentages** across different course components.

# Exams

- Time conflicts: Students who have time conflicts with other exams must notify the instructor about the conflict within one week of the start of the semester.

- Online exams: Exams may be held in the form of timed online exams and remote proctoring tools may be used. By signing up for this course, you confirm that you have read and accepted the terms and provisions of AUB's Privacy Statement.

# Late Assignment Submission Policy

- Students are expected to submit assignments on time by the due date specified for each assignment.
- Unless otherwise specified, students have the option of requesting a deadline extension and submitting assignments up to 48 hours late while incurring a 0.5% deduction per hour.
  - Students who wish to avail this option must request the deadline extension at least 24 hours prior to the submission deadline.

# Missed Exam Policy

- No makeup exams will be given for missed exams, except in the case of medical emergencies where the student presents a doctor’s note from AUBMC within one week after the exam date. If the student has a time conflict with other courses at AUB, the student must inform the instructor at the time the exam is announced so they can arrive at a reasonable accommodation. If the student fails to inform the instructor in a timely manner, the instructor is not obligated to make accommodations. Missed exams that are not made up will result in a zero grade for that exam.

# Regrading Policy

- If a student would like to request parts of an assignment or exam to be regraded, the student must make the regrading request within one week after the assignment or exam grade is released.

# Other Policies

- Please refer to the syllabus for other policies on:
  - Academic Integrity
  - Accessible Learning
  - Title IX, Non-Discrimination, and Anti-Harassment